

Attitude adjustment: Trends in the semantics of clausal-embedding

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Day 2

Truth values

Declarative sentences are objects that can be **true** or **false**.

- | | | | |
|-----|----|-------------------------|-------|
| (1) | a. | Debrecen is in Hungary. | true |
| | b. | London is in France. | false |

Yesterday, we didn't want to equate the meaning of *embedded* declaratives their **truth value**.

Do you remember why?

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Intuition:

We don't need to know whether a sentence is true or not to be able to understand it.

Something else goes into it, having to do with the ability to *check* whether it's true or false.

“To know the meaning of a (declarative) sentence is to know what the world would have to be like for the sentence to be true. Put another way, to give the meaning of a sentence is to specify its truth conditions, i.e., to give necessary and sufficient conditions for the truth of that sentence”

Dowty, Hall & Peters (1981)

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We use this to start modeling belief reports in a way that accounts for their descriptive properties.

Circumstances of evaluation

The truth value of a sentence *depends* on circumstances, for example, time and place.

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And there are expressions that 'shift' the circumstances of evaluation of a sentence.

- (5) a. **Tomorrow**, the Moon **will be** between the Sun and the Earth.
- b. **In England**, people drive on the left side of the road.

(slide presented on Aug. 11, 2026, before a solar eclipse)

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There are places (e.g., Sweden) where people have driven on the left side of the road prior to some point (H-Day) and then on the right following it. For this, it will depend on time as well.

Alternate worlds

The past, the future or other actual places are not the only cases of 'displacement'.

You'll often hear (or produce!) statements like:

- (6) a. **In a different world**, I have a house in the countryside and sell bread for a living.
- b. **There is an alternate universe in which** Japan won WWII.
- c. **It can't be otherwise.** Felix is home.

These sentences are true or false from our perspective, but they invite us to imagine **possibilities** different from ours.

Alternate possibilities

We really talk about different possibilities all the time and we barely ever signal it so explicitly. Modal verbs for example...

- (7) a. I **could** have a house in the countryside and sell bread for a living.
- b. Japan **could have** won WWII.
- c. Felix **has to** be home.

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Call these alternate possibilities **worlds**.

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For example, the person who asserts (8) is (usually) talking about the time and place at which they're located.

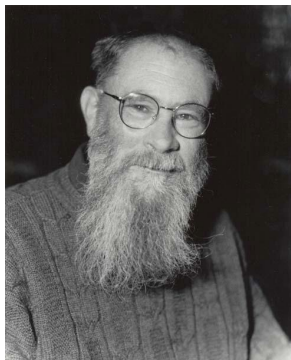
(8) People drive on the left side of the road.

But they are also talking about the *world* at which they're located, i.e., the *actual* world.

(Homework: Research why people drive on the side of the road that they do. Could it have been different?)

Before we move on, let's think about what possible worlds are.

Lewis, on possible worlds



Read at least the first two paragraphs of Lewis (1986), *On the Plurality of Worlds*.

1.1 The Thesis of Plurality of Worlds

The world we live in is a very inclusive thing. Every stick and every stone you have ever seen is part of it. And so are you and I. And so are the planet Earth, the solar system, the entire Milky Way, the remote galaxies we see through telescopes, and (if there are such things) all the bits of empty space between the stars and galaxies. There is nothing so far away from us as not to be part of our world. Anything at any distance at all is to be included. Likewise the world is inclusive in time. No long-gone ancient Romans, no long-gone pterodactyls, no long-gone primordial clouds of plasma are too far in the past, nor are the dead dark stars too far in the future, to be part of this same world. Maybe, as I myself think, the world is a big physical object; or maybe some parts of it are entelechies or spirits or auras or deities or other things unknown to physics. But nothing is so alien in kind as not to be part of our world, provided only that it does exist at some distance and direction from here, or at some time before or after or simultaneous with now.

The way things are, at its most inclusive, means the way this entire world is. But things might have been different, in ever so many ways. This book of mine might have been finished on schedule. Or, had I not been such a commonsensical chap, I might be defending not only a plurality of possible worlds, but also a plurality of impossible worlds, whereof you speak truly by contradicting yourself. Or I might not have existed at all – neither I myself, nor any counterpart of me. Or there might never have been any people. Or the physical constants might have had somewhat different values, incompatible with the emergence of life. Or there might have been altogether different laws of nature; and instead of electrons and quarks, there might have been alien particles, without charge or mass or spin but with alien physical properties that nothing in this world shares. There are ever so many ways that a world might be; and one of these many ways is the way that this world is.

Take the following sentence and its truth conditions:

(9) The sentence “It’s raining” is true at w iff it’s raining at w .

Assume that Ω is the set of all possible worlds.

One can imagine multiple distinct worlds at which the sentence is true and collect them in a set.

(10) $\llbracket \text{it's raining} \rrbracket = \{w : \text{it's raining at } w\} =$
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Propositions are objects of type $\langle s, t \rangle$.

To check whether a proposition is true or not at an arbitrary world w_{18} , we check whether that world is a member of the proposition:

(11) Is it the case that $w_{18} \in \llbracket \text{it's raining} \rrbracket$

If yes, the proposition is true at that world; if not, it is false at that world.

a picture with two propositions

Bruno is in
Austria
..

Bruno is not
in Austria
..

it's raining:

w_1

w_2

it's not raining: w_3

w_4

For declarative sentences:

Their *extension* (at a world) is a truth value,
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Other expressions have intensions and extensions as well (just to illustrate that declarative sentences aren't special):

“The Prime Minister of Hungary”

 extension: Péter Magyar

 intension: a function ($\langle s, e \rangle$) from worlds to the person who holds the office at that world

Outline

The Hintikka idea

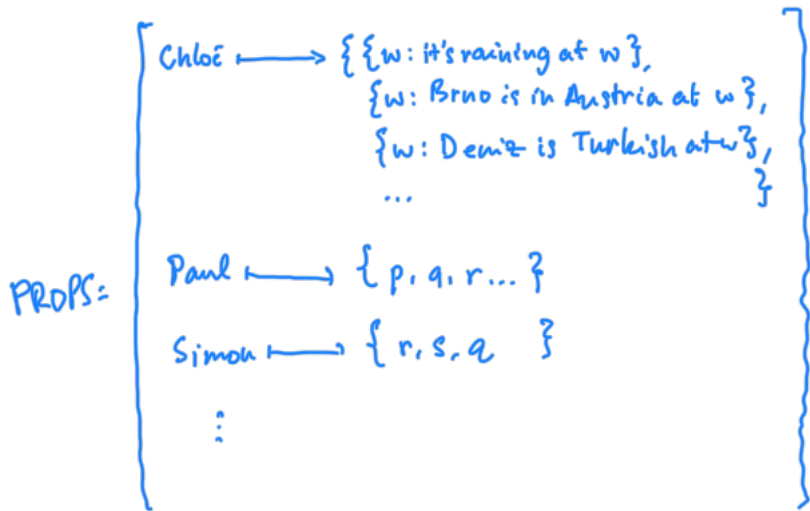
1. The truth conditions of belief reports
2. How they are compositionally derived

We can use possible worlds to better model the meaning of declarative sentences. But how do we make the link with belief?

People believe propositions, and more than one of them.

We can define functions between individuals and the sets of propositions that they believe.

A map between individuals and sets of propositions



Call this function PROPS.

(12) PROPS(Chloé)
= {it's raining, Brno is in Austria, Deniz is Turkish, ...}

The most immediate way of checking whether a sentence like 'Chloe thinks that it's raining' is to check whether:

(13) $\llbracket \text{that it's raining} \rrbracket \in \text{PROPS}(\text{Chloé})$

If yes, the sentence is true; if not, it is false.

Because it makes our lives easier (?), we usually don't work with PROPS directly, but with its grand intersection.

Refresher on intersection

Intersection for two sets

$$(14) \quad \{1, 2, a, b, c\} \cap \{a, b, c\} = \{a, b, c\}$$

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Grand intersection for a set of sets

$$(15) \quad \bigcap \{A, B, \dots, Z\} = A \cap B \cap \dots \cap Z$$

$$(16) \quad \bigcap \{\{1, 2, a, b, c\}, \{a, b, c\}, \{a, c\}\} = \{a, c\}$$

$$(17) \quad \bigcap \text{PROPS}(\text{Chloé}) = \{p, q, r, \dots\} = p \cap q \cap r \dots$$

Call this set $\text{DOX}(\text{Chloé})$. It corresponds to the set of all *possible worlds* in which all of the propositions that Chloé believes are true.

$$(18) \quad \begin{aligned} \text{DOX}(\text{Chloé}) &= \bigcap \text{PROPS}(\text{Chloé}) \\ &= \{w : w \text{ is compatible with Chloé's beliefs} \} \end{aligned}$$

Now, checking whether Chloé believes some proposition p , we check whether (19) is true:

$$(19) \quad \text{DOX}(\text{Chloé}) \subseteq \llbracket \text{that it's raining} \rrbracket \\ \Leftrightarrow \quad \text{DOX}(\text{Chloé}) \subseteq \{w : \text{it's raining at } w\}$$

As Hintikka puts it



"My basic assumption (slightly oversimplified) is that an attribution of any propositional attitude to the person in question involves a division of all the possible worlds (more precisely, all the possible worlds which we can distinguish in the part of language we use in making the attribution) into two classes: into those possible worlds which are in accordance with the attitude in question and into those which are incompatible with it."

Questions for you

What would it mean for $\text{DOX}(\text{Chloé})$ to be the singleton set containing the actual world?

What would it mean for $\text{DOX}(\text{Chloé})$ to be empty?

What would it mean for $\text{DOX}(\text{Chloé})$ to be equal to the set of all possible worlds Ω ?

Why didn't we opt for

$$(20) \quad \text{DOX}(\text{Chloé}) = p$$

These global truth conditions for belief reports are traditionally presented together with a syntactic hypothesis.

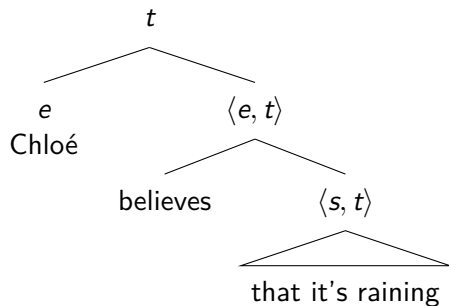
In the trees below, what's important is the assumption that:

the clause is the direct object of 'believe' syntax

the denotation of the clause is the argument of the function
denoted by 'believe' semantics

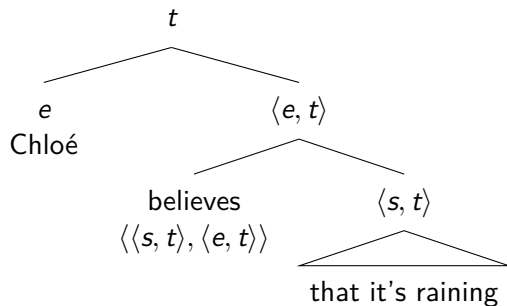
Getting to these truth conditions compositionally

types



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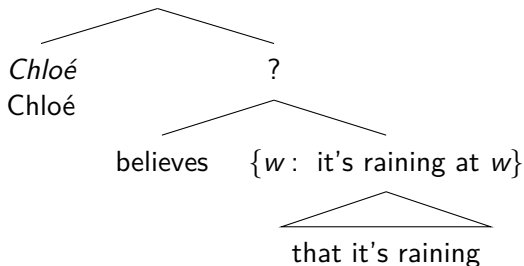
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Getting to these truth conditions compositionally

the function

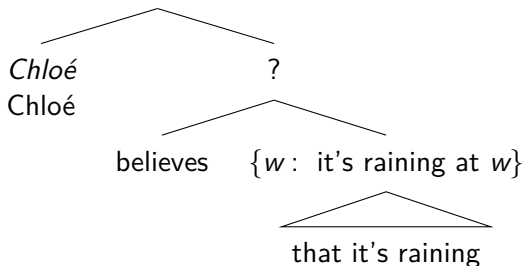
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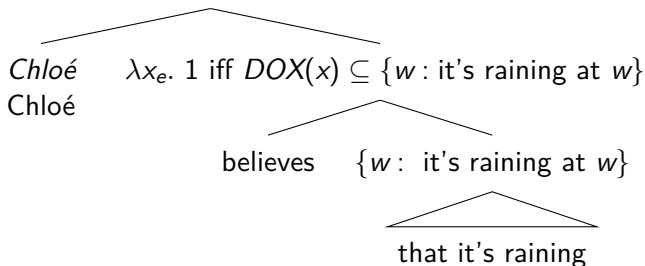


$\llbracket \text{believes} \rrbracket = \lambda p_{\langle s, t \rangle} . \lambda x_e . 1 \text{ 1 iff } \text{DOX}(x) \subseteq p$

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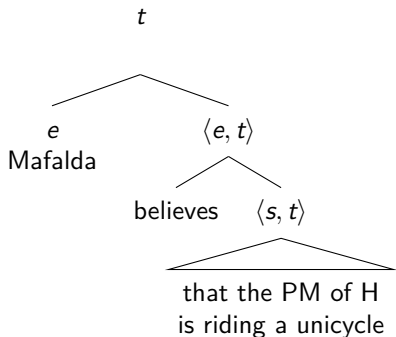
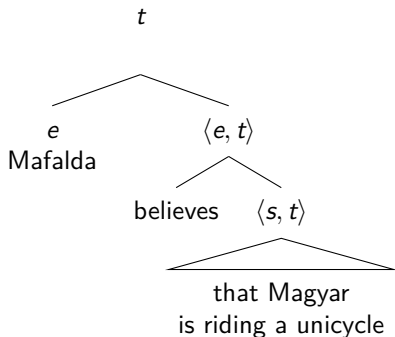
One of our problems yesterday was that if 'believe' were a function of type $\langle t, \langle e, t \rangle \rangle$, it would map two sentences with the same truth value on to the same output.

While these two sentences are true at our world (now), they do not have to have the same intension—they correspond to two distinct propositions.

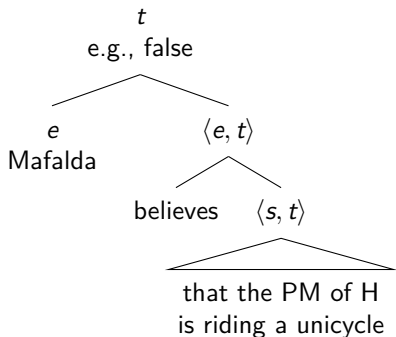
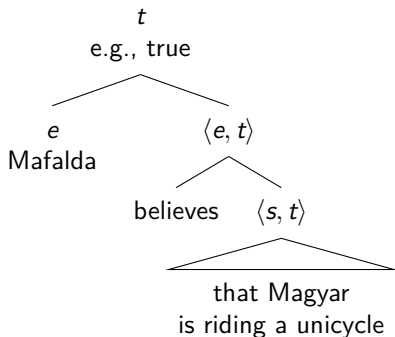
- (21) a. Péter Magyar is riding a unicycle
 b. The Prime Minister of Hungary is riding a unicycle.

This is because there are worlds at which the PM of Hungary is Magyar, but others at which it is David Müller, Ari Joshi, my sister, etc.

The two propositions in the trees below are *different* sets of worlds.
'Believes' need not map onto the same object.



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Going back to the jackalope example, we can imagine that there are worlds at which there are jackalopes.

And sentence (22) is false, but it could be that all of *Mafalda's belief worlds* have jackalopes in them.

(22) Jackalope sentence

If you're interested in further details, some places to start:

Heim & Kratzer (1998), chapter 2.

von Fintel & Heim (2011), chapters 1 & 2.